UAM-MARL

Uncertainty-Aware Modality-Enhanced Multi-Agent RL

with LLM-Guided Graph Policies

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Background and Motivation

Widespread Applications of MAS

Core Application Scenarios

Covering home smart assistants, automated warehousing logistics, post-disaster search and rescue missions, and long-term monitoring in complex environments.

Core Challenges

1. Collaboration efficiency and resource allocation in dynamic environments
2. Reward design for aligning local behaviors with global goals
3. Communication synchronization and system robustness in complex tasks

Limitations of Current Methods

Limitations of LLM-driven MARL

Mainstream methods rely heavily on the strong assumption of "perfect perception," ignoring noise and sensor errors in real environments.

Semantic-Perception Gap

There is a huge deviation between the abstract semantic plans generated by language models and the low-quality perceptual data obtained by agents from the environment, leading to task failure.

Motivation of This Study

Bridging the Perception Gap

Enable LLM-driven agents to maintain reliable collaboration and task execution capabilities in noisy, occluded, and non-ideal environments.

Goal: Enhance System Robustness

Core Challenges

Perceptual Uncertainty

Visual detection may misclassify objects, and sensors are prone to data loss or environmental noise interference.

Decisions based on incorrect perception trigger chain reactions, leading to task failure.

Mismatch between Plan and Reality

Abstract plans from LLMs often ignore real-time states and available resources.

Agents waste resources executing unfeasible subtasks, causing process stagnation.

Misleading Reward Signals

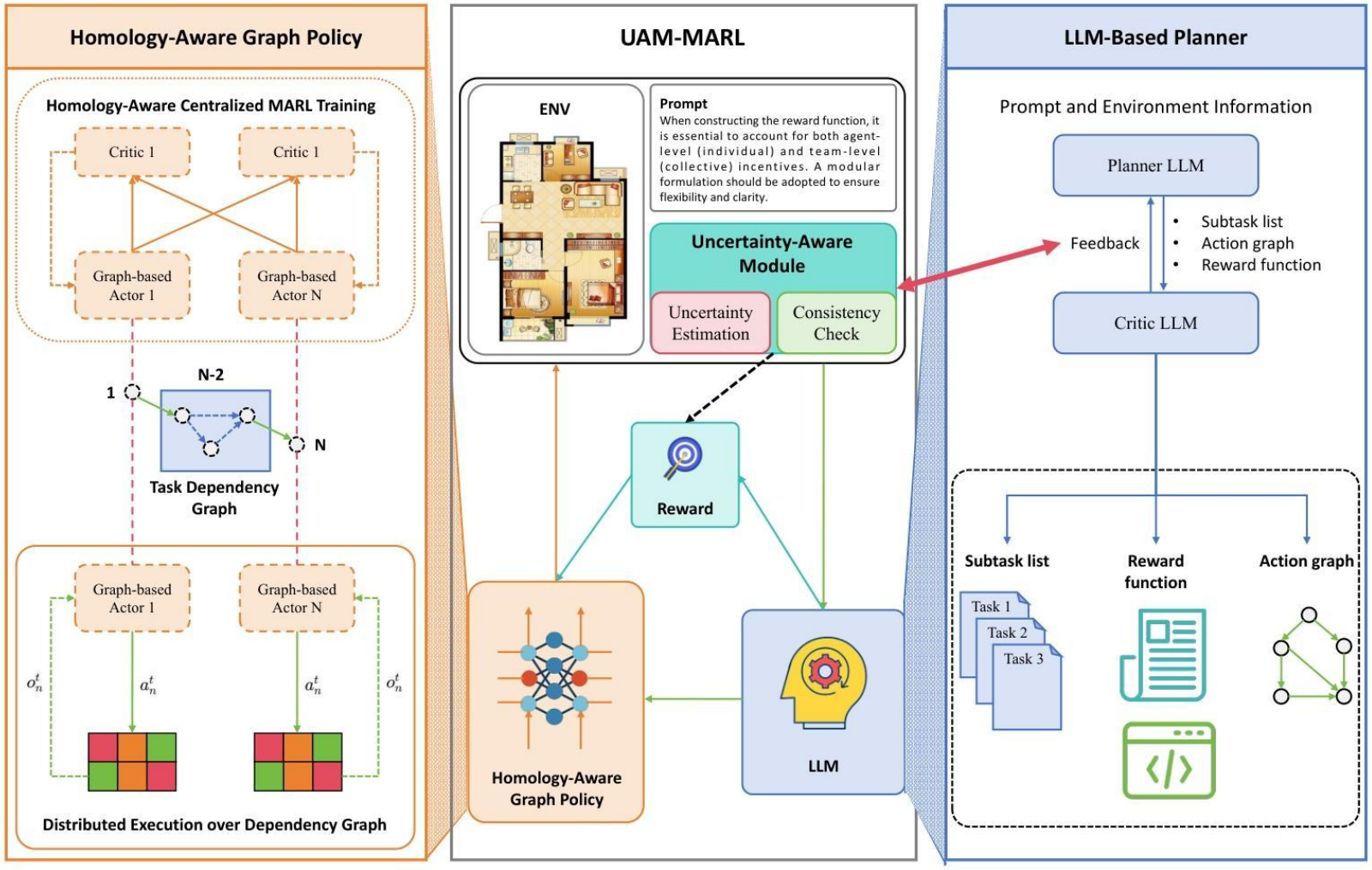
Traditional rewards fail to distinguish confidence levels, rewarding "false achievements".

Reinforcement learning is misled by noise, leading to poor generalization.

Inefficient Coordination Structure

Redundant cycles and dependencies in task decomposition increase overhead.

Poor team coordination and resource scheduling hinder scalability.


Homology-Aware Graph Policy
Homology-Aware Centralized MARL Training
Critic 1
Critic 1
Graph-based Actor 1
Graph-based Actor N
N-2
1
N
Task Dependency Graph
Graph-based Actor 1
Graph-based Actor N
Distributed Execution over Dependency Graph
UAM-MARL
ENV
Prompt
When constructing the reward function, it is essential to account for both agent-level (individual) and team-level (collective) incentives. A modular formulation should be adopted to ensure flexibility and clarity.
Uncertainty-Aware Module
Uncertainty Estimation
Consistency Check
Reward
Homology-Aware Graph Policy
LLM
LLM-Based Planner
Prompt and Environment Information
Planner LLM
Feedback
Subtask list
Action graph
Reward function
Critic LLM
Subtask list
Reward function
Action graph
Task 1
Task 2
Task 3

LLM Planner-Critic

From Natural Language Instructions to Executable Task Plans

STEP 01 / Instruction Input

Receive high-level natural language instructions, such as "Clean the dining table and take out the trash", as the system's start signal.

STEP 02 / Planner (Planner LLM)

Decompose the instruction into an atomic subtask sequence and construct the initial task dependency graph G0 based on task logic.

STEP 03 / Critic (Critic LLM)

Act as an "inspector", evaluate the logical rationality of G0, identify contradictions or infeasible steps, and feed back to the planner for iterative correction.

STEP 04 / Reliable Plan Output

Output the final task dependency graph that has been verified and corrected through multiple rounds, which is logically rigorous and executable, to guide the downstream executor.

Core Advantage: Through the closed-loop mechanism of "Planning-Criticism-Iteration", it significantly suppresses the "hallucination" problem of LLM, ensuring that the generated task plans are highly executable and robust.

Homology-Guided Graph Policy

Building Efficient Collaboration Networks with Topology

Core Pain Point: Redundant Cycles

The task dependency graph contains numerous redundant cycles, leading to inefficient coordination.

Graph Topology Analysis

Mathematically interpret the first original task dependency graph the G as a **Simplicial Complex** to establish that a topological foundation.

Independent Cycle Detection

Introduce **First-order Homology** theory to algorithmically detect and identify all independent cycles in the graph.

Core Advantages of the Strategy

- Cost Reduction & Efficiency
- Focus on Key Points

Redundant Cycle Pruning

Calculate cycle correlation scores, set thresholds, dynamically remove low-correlation redundant edges, and retain core collaboration links.

Efficient Policy Generation

Based on the adjacency matrix M of the pruned sparse graph G, guide the Graph Neural Network (GNN) to learn the optimal collaboration strategy.

Core Logic: Topology Analysis ➔ Cycle Detection ➔ Redundant Pruning ➔ GNN Policy Generation

Experimental Setup

Platform and Task Configuration

- Simulation Platform:AI2-THOR(High-fidelity interactive 3D home environment simulator)
- Core Tasks: Object search, delivery, and manipulation (4-6 agents)

Real-World Robustness Test (Noise Injection)

Visual Misclass
Category Bias

Object Occlusion
Partial/Full

Modality Loss
Visual/Lang

Core Evaluation Metrics

- Success Rate
- Avg. Completion Time
- Token Cost & Conflict Count
- Uncertainty Robustness

$UR = SR_n / SR_0$

AI2-THOR Multi-Scene Simulation Environment

Provides diverse 3D home scenes including kitchen, living room, bedroom, and bathroom. Supports physical interaction and environment exploration, serving as a standard testbed for multi-agent collaboration algorithms.

Main Results: Performance Comparison

Detailed Comparison of Experimental Data in Scene 1

对比方法	成功率	耗时(s)	Token
Centralized LLM	0.59	99.59	6.3
LLMs Dialog	0.67	154.54	10.6
UAM-MARL (Ours)	0.95	62.36	0.9

Significantly Leading Success Rate

Achieves the highest success rate of 0.95, significantly outperforming baseline methods like Centralized LLM.

Optimal Task Execution Efficiency

With an average time of only 62.36s, it is significantly faster than other comparison groups.

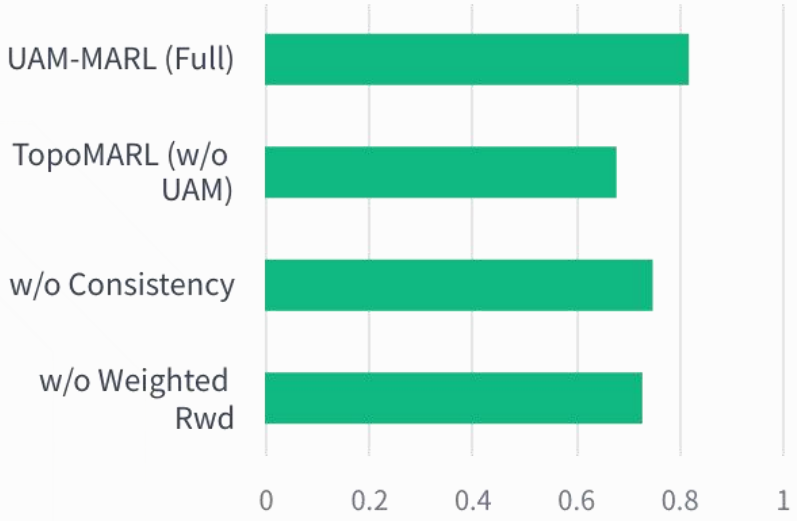
Lowest Token Calculation Cost

Consumes only 0.9 Tokens, approximately 1/7 to 1/10 of other methods, offering great economic benefits.

Conclusion: UAM-MARL demonstrates overwhelming advantages in task completion ability, execution efficiency, and calculation cost, fully verifying the effectiveness of the algorithm.

Main Results: Robustness Analysis

Comparison of Uncertainty Robustness (UR) Across Models



Core Metric: Uncertainty Robustness (UR)

A UR value closer to 1 indicates that the model suffers less performance loss under noisy environments, meaning the system has stronger anti-interference ability (robustness).

Key Experimental Data Insights

- The full model UAM-MARL achieved the best performance with UR = 0.82.
- After removing the UAM module, UR dropped sharply to 0.68, and all ablation experiments showed significant decreases.

Conclusion: Uncertainty Awareness is Key

The uncertainty modeling design of UAM-MARL enables it to effectively handle perceptual noise in real environments, which is a key reason for achieving superior performance.

Ablation Study

Component Removal Strategy	Task Success Rate	Avg. Conflicts	Avg. Time (ms)
Full Model (Baseline)	0.91 (Best)	3.2 (Lowest)	74.6
Remove Uncertainty Module	↓ 0.83 (Significant Drop)	↑ 5.1	-
Remove Consistency Check	-	↑ 4.2	-
Remove Weighted Reward / Homology Graph	-	↑ 4.8	↑ 84.5 (Efficiency Drop)

Conclusion: The Whole Framework is Greater Than the Sum of Its Parts

Each core component of UAM-MARL (Uncertainty Module, Consistency Check, Weighted Reward, etc.) makes a significant and indispensable contribution to the final superior performance. The absence of a single component directly leads to a drop in success rate or a surge in conflicts, verifying the necessity of multi-module collaboration.

Conclusion and Future Work

Core Conclusions

Proposing the UAM-MARL Framework

To address the "semantic-perceptual gap," we constructed a unified multi-agent reinforcement learning framework that organically integrates high-level semantics with low-level perception.

Core Innovation Achieved

Integrating uncertainty awareness and cross-modal consistency checks, designing a weighted reward mechanism, and combining LLM planning with topological strategies significantly enhance robustness.

Multi-dimensional Experimental Verification

In the AI2-THOR environment, our method outperforms SOTA approaches in task success rate, execution efficiency, resource cost.

Future Work

Scaling to Large-Scale Agent Swarms

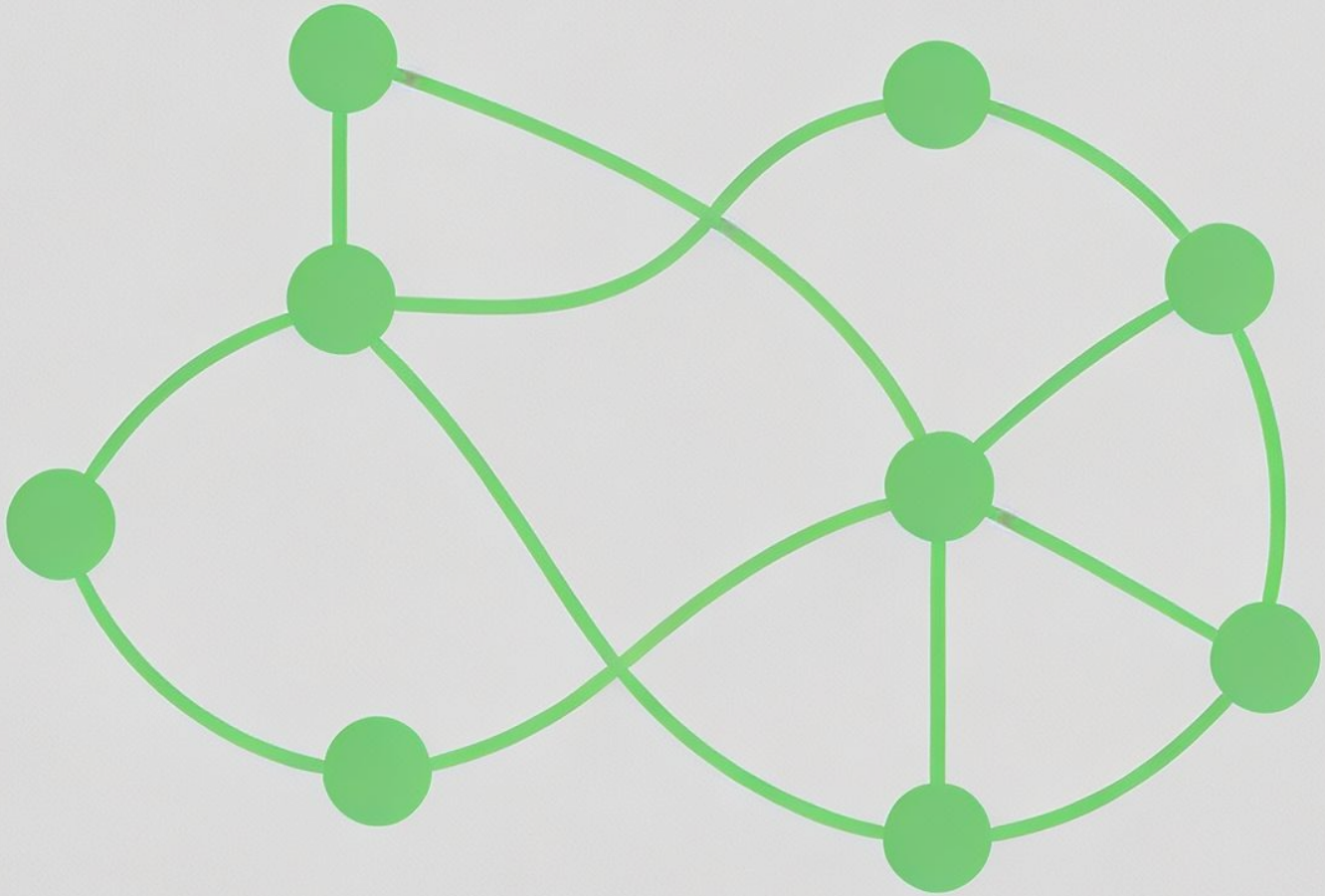
Breaking the current limit of small-scale teams, exploring scheduling strategies and communication efficiency optimization in tasks with hundreds of agents.

Real-World Physical System Validation

Transferring algorithms from simulation to real robot platforms to validate actual execution in dynamic unstructured scenarios.

Deepening Interdisciplinary Research

Deeply exploring the underlying integration logic of language reasoning, topological space analysis, and uncertainty modeling in reinforcement learning.



Thank You!